Convergence Simulations Model A

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1 Model Structure

1.1 Data and indexing

The data are N binary trials. Trial i has

- a response $y_i \in \{0, 1\}$, the monkey's categorization of the stimulus as "low" (0) or "high" (1);
- a standardized stimulus amplitude x_i. The raw amplitudes are centered and scaled by the mean and standard deviation of the distinct amplitude levels. The covariate row is $(1, x_i)$;
- a group $g[i] \in \{1, \ldots, 4\}$: the combination of hand and distractor condition,

$$g \in \{\text{left_uni},\ \text{left_bi},\ \text{right_uni},\ \text{right_bi}\},$$

  where "uni" is unimanual (no distractor) and "bi" is bimanual (distractor present).

Model A ignores sessions: all trials of a group are pooled, as if every session had the same psychometric curve. The real data have $N = 12{,}937$ trials, about 3,230 per group.

1.2 Likelihood

Each response is Bernoulli,

$$y_i \sim \text{Bernoulli}(\psi_i), \qquad \psi_i = \psi\big(x_i;\ \theta_{g[i]}\big),$$

with the psychometric function

$$\psi(x; \theta) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x), \qquad \theta = (\beta_0, \beta_1, \gamma_h, \gamma_l).$$

Here β_0 and β_1 are parameters of the sigmoid, and γ_h, γ_l set the asymptotes:

$$\lim_{x \to -\infty} \psi = \gamma_h, \qquad \lim_{x \to +\infty} \psi = 1 - \gamma_l.$$

So γ_h is the probability of responding 1 ("high") at very low amplitudes, and γ_l the probability of responding 0 ("low") at very high amplitudes. In the code ψ_i is clipped to $[10^{-8}, 1 - 10^{-8}]$ before entering the Bernoulli likelihood, for numerical stability.

1.3 Priors

Each group has its own four parameters, independent of the other groups, with the same priors in every group:

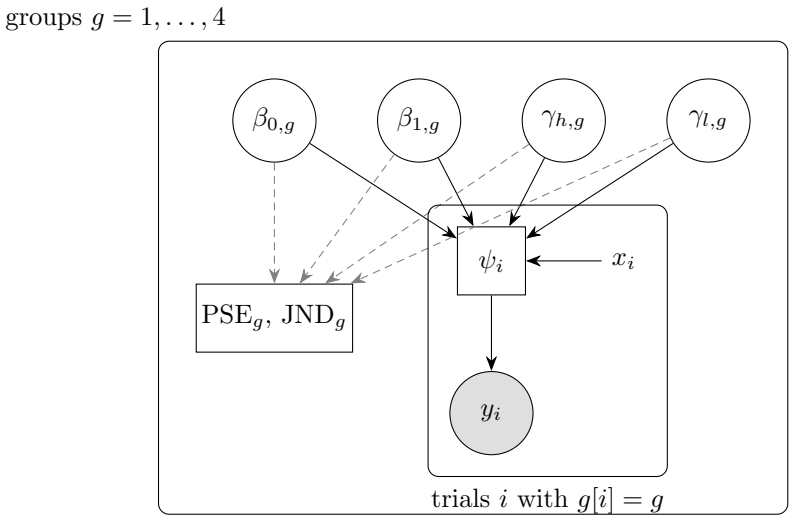
$$\begin{aligned}
\beta_{0,g} &\sim \mathcal{N}(0,\ 6), \\
\beta_{1,g} &\sim \mathcal{N}(4,\ 2), \\
\gamma_{h,g} &= 0.25\, w_{h,g}, \qquad w_{h,g} \sim \text{Beta}(\text{mean } 0.2,\ \text{sd } 0.15), \\
\gamma_{l,g} &= 0.25\, w_{l,g}, \qquad w_{l,g} \sim \text{Beta}(\text{mean } 0.2,\ \text{sd } 0.15).
\end{aligned}$$

The Beta distribution has shape parameters $a = 1.22$, $b = 4.89$. Scaling by 0.25 keeps each lapse parameter in $(0, 0.25)$, so the curve is increasing and the PSE and JND below are well defined.

These priors have not been checked by prior predictive simulation. What they imply:

- β_0: very wide. Together with the slope prior it puts the PSE inside the stimulus range (6 to 50 μm) only 64% of the time.
- β_1: a symmetric normal, so 2.3% of the prior mass is on negative (decreasing) slopes.
- γ_h, γ_l: mean 0.05, median 0.041, 95% range [0.003, 0.141]. This is centered well above the lapse rates of trained animals (often below 0.01), which matters for γ_l (Section 3.3).

1.4 Graphical Model



Circles are random variables (shaded: observed), rectangles are deterministic functions, and x_i is a fixed covariate. Dashed arrows lead to derived quantities that do not feed back into the likelihood.

1.5 Derived Quantities: PSE and JND

Point of subjective equality (PSE): the stimulus at which a response of "high" and "low" are equally likely ($\psi = 1/2$),

$$\text{PSE} = \frac{1}{\beta_1}\left[-\beta_0 + \log\frac{1-2\gamma_h}{1-2\gamma_l}\right].$$

Just noticeable difference (JND): half the stimulus distance between the points where $\psi = 0.25$ and $\psi = 0.75$,

$$\text{JND} = \frac{x_{.75} - x_{.25}}{2} = \frac{1}{2\beta_1} \log \frac{(3 - 4\gamma_h)(3 - 4\gamma_l)}{(1 - 4\gamma_h)(1 - 4\gamma_l)}.$$

Both are in units of the standardized stimulus; one unit is 14.6 μm.

1.6 Implementation

The amplitudes are standardized, and β_0, β_1 are written as deterministic transforms of standard normal variables. The model is fit with NUTS, 4 chains of 1000 tuning and 1000 kept draws.

2 Synthetic Data Generation

`Convergence_A.py` fixes one set of true parameters and fits Model A to synthetic datasets of increasing size:

	left_uni	left_bi	right_uni	right_bi
β_0	-1	1	-1	1
β_1	3	3	6	6
γ_h	0.03	0.06	0.06	0.03
γ_l	0.005	0.01	0.005	0.01
PSE	0.316	-0.369	0.147	-0.174
JND	0.383	0.403	0.200	0.193

- **Trials.** Each trial's stimulus level and group are drawn with replacement from the real experimental design, so every dataset keeps the real mix of the 8 stimulus levels and 4 groups. Responses are then simulated from the true curves.
- **Sizes.** 250, 500, 1000, 2000, 4000, 8000 and 16,000 trials in total (about 60 to 4,000 per group). The real data, with 12,937 trials, lie between the two largest sizes.
- **Nested datasets.** Within a replicate, the size-N dataset is the first N trials of one long simulated run, so each larger dataset only adds trials to the smaller one. The trend with size is then not mixed with the noise of drawing a new dataset at every size.
- **Replicates.** The whole sweep was repeated 20 times with independent simulated data (140 fits). Each size and parameter therefore has $20 \times 4 = 80$ fits from which to estimate coverage. A calibrated model would give a coverage between 0.90 and 0.99 in 95% of such cells, and about 0.95 on average.

An earlier run used only 3 replicates (12 fits per cell). Its coverage of the PSE dropped to 0.67–0.75 between 1,000 and 4,000 trials, almost entirely because of one replicate whose left_uni PSE stayed about 3 posterior sds above the truth. With 20 replicates that pattern does not recur (see below), so it was chance in one simulated dataset.

3 Results

Every figure and number in this report is produced by `Convergence_A.py`; the cell "Numbers for Convergence_A_Report.tex" prints the numbers that the earlier cells do not, and also runs on the saved results without refitting.

3.1 Sampling diagnostics

All 140 fits sampled well: $\hat{R} \leq 1.006$, a bulk ESS of at least 1,554 for every parameter, and a single divergence in total (one fit at 1,000 trials).

3.2 Coverage

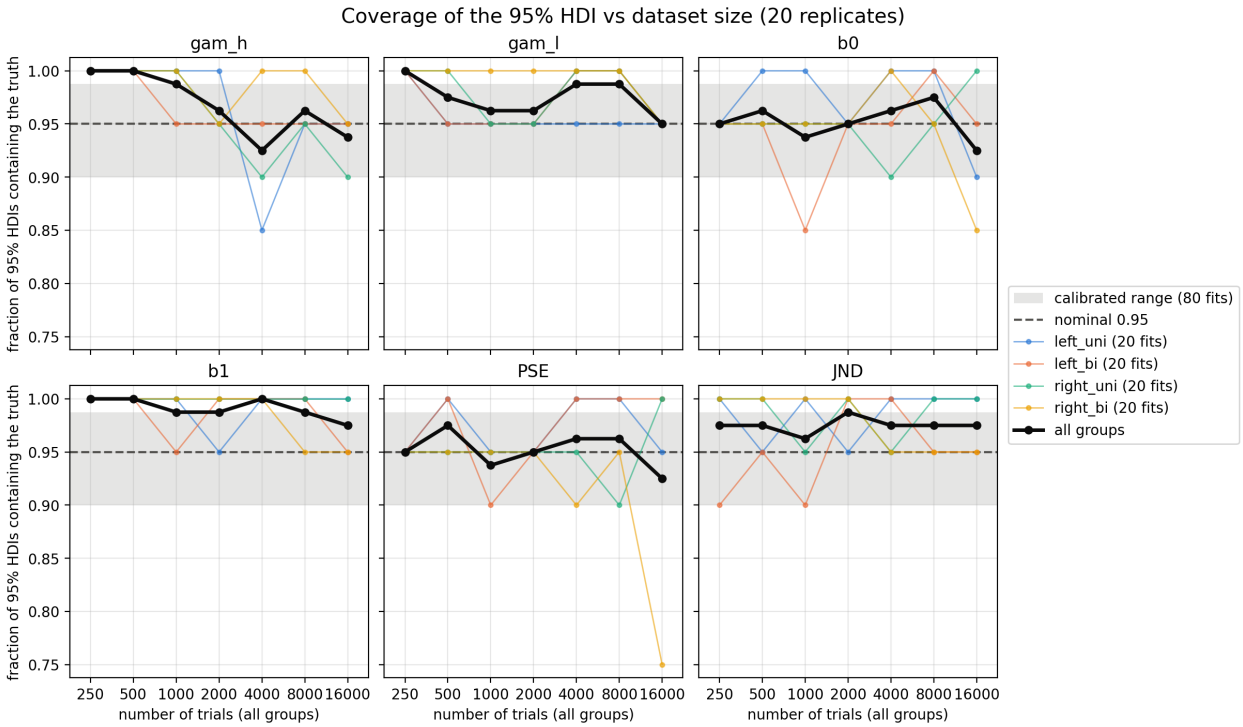


Figure 1: Fraction of 95% HDIs that contain the true value. Black: all groups (80 fits per point), with the range expected from a calibrated model in gray. Colored: each group separately (20 fits per point, so they vary much more).

Table 1: Coverage of the 95% HDI over 80 fits (20 replicates × 4 groups).

Trials	250	500	1000	2000	4000	8000	16000
γ_h	1.00	1.00	0.99	0.96	0.92	0.96	0.94
γ_l	1.00	0.98	0.96	0.96	0.99	0.99	0.95
β_0	0.95	0.96	0.94	0.95	0.96	0.98	0.92
β_1	1.00	1.00	0.99	0.99	1.00	0.99	0.98
PSE	0.95	0.98	0.94	0.95	0.96	0.96	0.92
JND	0.98	0.98	0.96	0.99	0.98	0.98	0.98

Every coverage value lies in the calibrated range 0.90–0.99 (Table 1), except for a few values of 1.00 (γ_h, γ_l and β_1 at the smallest sizes, and β_1 at 4,000 trials): there the HDIs are wider than needed, because the prior still contributes. The 95% HDIs of Model A can therefore be taken at face value at the size of the real data.

Within single groups (20 fits per value) one value stands out: the coverage of the right_bi PSE at 16,000 trials is 0.75 (15 of 20), with β_0 for the same group at 0.85. With 168 group-level coverage values, one this low can happen by chance, and the same group is at 0.95 at 8,000 trials. But at 16,000 trials its PSE error (RMSE 0.018) is somewhat larger than its posterior sd (0.014), so there may be a mild undercoverage of the PSE for steep curves at large sizes. At a slope of 6 the curve rises over less than one stimulus spacing, so only the two levels around the PSE carry information about it.

3.3 Bias

Table 2: Average error of the posterior mean in units of the posterior sd (mean of (mean − true)/sd over 80 fits). 0 is unbiased.

Trials	250	500	1000	2000	4000	8000	16000
γ_h	0.07	0.03	−0.04	−0.02	−0.11	−0.03	0.06
γ_l	1.11	1.07	0.98	0.84	0.69	0.51	0.36
β_0	0.11	0.16	0.11	0.11	0.06	0.05	−0.03
β_1	0.41	0.40	0.41	0.32	0.23	0.22	0.17
PSE	−0.02	−0.05	−0.07	−0.07	−0.10	−0.08	−0.04
JND	−0.16	−0.20	−0.28	−0.19	−0.12	−0.11	−0.08

PSE, JND, β_0 and γ_h are recovered without meaningful bias (within about 0.3 posterior sd at every size, Table 2). Only 7 of the 3,360 parameter estimates (0.2%) were more than 3 posterior sds from the truth, as expected by chance.

γ_l is estimated too high. The true values (0.005 and 0.01) lie in the lower tail of the prior, whose mean is 0.05, and the data contain very few lapses of this kind: at 16,000 trials a group with $\gamma_l = 0.005$ has about 4,000 trials, but only 13% (left_uni) to 37% (right_uni) of them are at levels where the curve is near its ceiling (above 0.95), so the data contain only about 3 to 7 such lapses. The prior therefore pulls the estimates up, by about one posterior sd at small sizes, decreasing to 0.36 sd at 16,000 trials (Figure 2). The posteriors are wide enough that coverage stays at nominal.

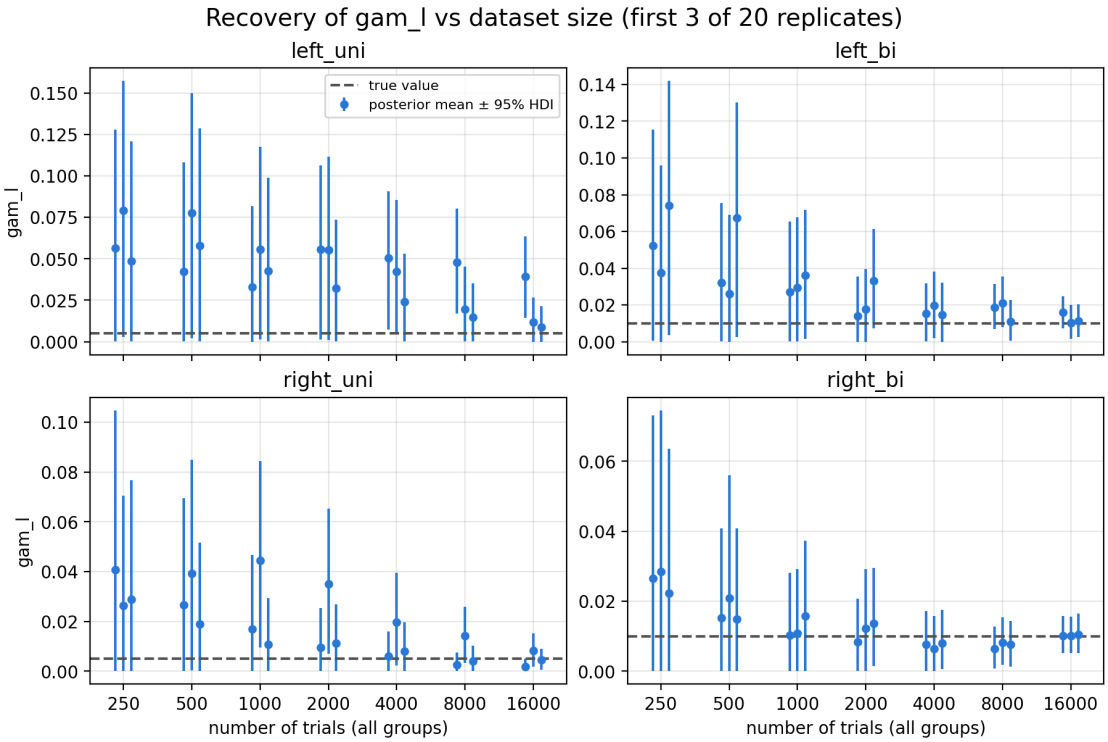


Figure 2: Posterior mean and 95% HDI of γ_l against dataset size, for the first 3 of the 20 replicates. Dashed: true value.

left_uni: γ_l and slope trade off. The bias is largest and most persistent for left_uni: at 16,000 trials its γ_l averages 0.015 against a true 0.005 (still 1.1 posterior sds too high), and its slope β_1 is 0.12 too high (0.6 sd). The left_uni curve only reaches about 0.97 at the highest stimulus, so the data cannot tell whether the remaining shortfall is a lapse (γ_l) or a curve that has not yet reached its ceiling (the slope), and a higher γ_l is compensated by a steeper slope. For the other three groups the γ_l bias falls to 0.3 sd or less by 16,000 trials. The slope bias at small sizes (1.3–1.5 too high at 250 trials for both left-hand groups) comes from the prior mean of 4 pulling the true slope of 3 upward.

3.4 Precision

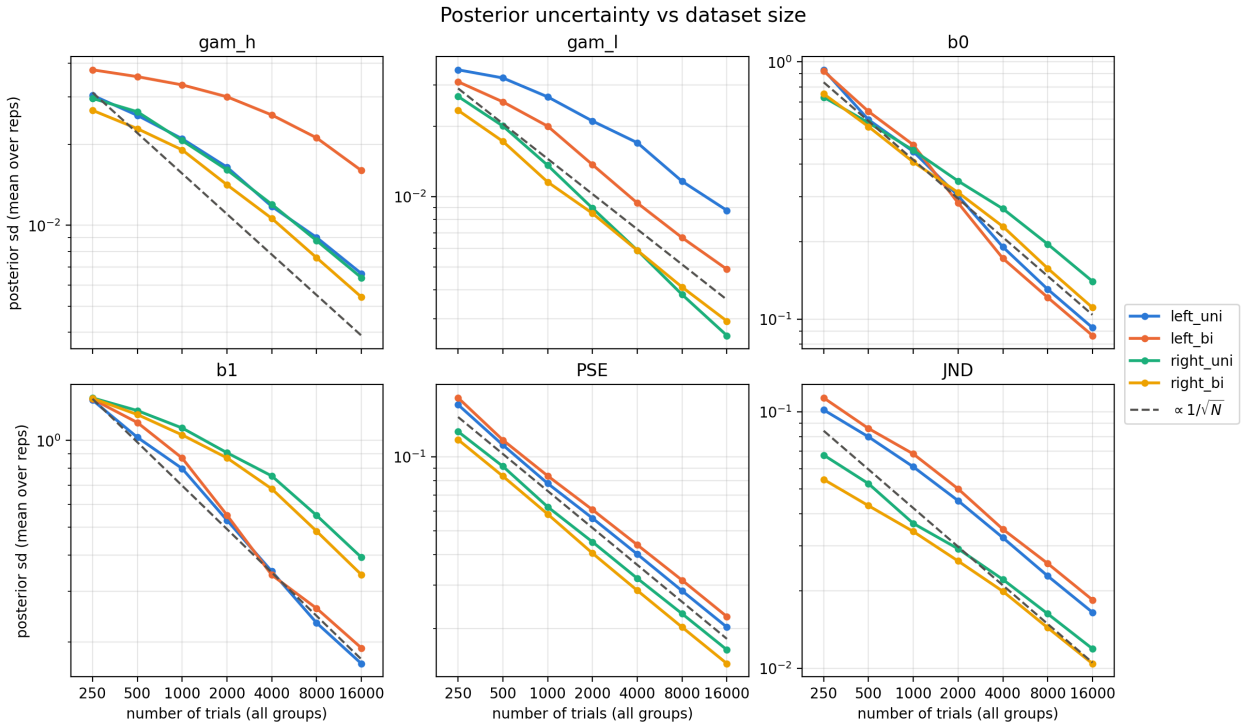


Figure 3: Posterior sd against dataset size (log-log), averaged over replicates. The dashed line has the slope $-1/2$ expected once the data dominate.

Table 3: Posterior sd, averaged over groups and replicates (standardized units for PSE and JND).

Trials	250	500	1000	2000	4000	8000	16000
γ_h	0.031	0.028	0.023	0.019	0.015	0.012	0.009
γ_l	0.029	0.024	0.018	0.013	0.010	0.007	0.005
β_0	0.83	0.59	0.45	0.31	0.22	0.15	0.11
β_1	1.39	1.17	0.95	0.71	0.53	0.38	0.27
PSE	0.145	0.101	0.071	0.051	0.036	0.026	0.018
JND	0.084	0.065	0.050	0.038	0.027	0.020	0.014

All posteriors narrow steadily with more data (Figure 3, Table 3). PSE and β_0 narrow at the $1/\sqrt{N}$ rate over the whole range, so for them the data dominate from 250 trials on. β_1, JND and the lapse rates narrow more slowly at small sizes, where the prior still contributes. β_1, JND and γ_l reach the $1/\sqrt{N}$ rate from about 4,000 trials; γ_h is still a little slower at 16,000.

At 16,000 trials the posterior sd of the PSE is 0.018 standardized units (0.27 μm) and that of the JND 0.014 (0.21 μm). Scaled to the real data (about 0.8 times as many trials per group) these become about 0.30 and 0.23 μm.

3.5 Psychometric curves

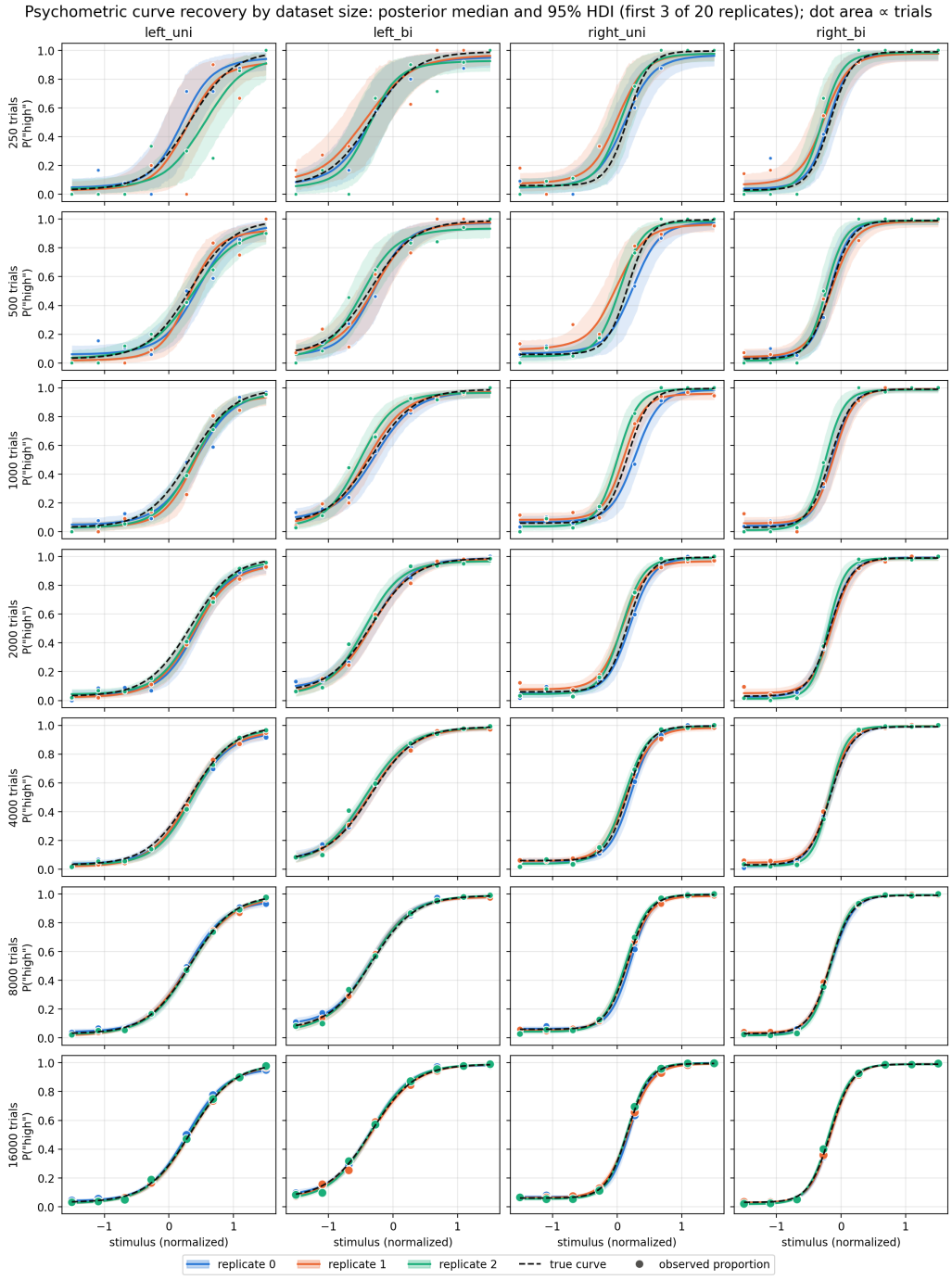


Figure 4: Recovered psychometric curves (posterior median and pointwise 95% HDI) at each size, for the first 3 replicates, with the true curve dashed and the observed proportions as dots.

Table 4: Curve-level recovery, averaged over groups and replicates: mean width of the pointwise 95% HDI of ψ, largest absolute error of the posterior median curve, and fraction of the stimulus range where the true curve lies inside the HDI band.

Trials	250	500	1000	2000	4000	8000	16000
HDI width	0.226	0.172	0.129	0.095	0.068	0.049	0.035
largest error	0.126	0.100	0.077	0.054	0.042	0.027	0.021
truth in band	0.97	0.97	0.95	0.97	0.96	0.96	0.93

The whole curve is recovered well (Figure 4, Table 4): at 16,000 trials the HDI band is on average 0.035 wide in ψ, the median curve is never more than about 0.02 from the truth, and the true curve lies inside the band over 93% of the stimulus range.

4 Conclusions for the real data

At the size of the real data (12,937 trials, about 3,230 per group):

- **Reliable:** the PSE and JND of every group, β_0, and γ_h: unbiased, with calibrated 95% HDIs. The posterior sd is about 0.3 μm for the PSE and 0.2 μm for the JND.
- **Reliable but with conservative HDIs:** β_1; its coverage is 0.98–0.99.
- **Biased upward:** γ_l, by roughly 0.4–0.5 posterior sd overall, and by about one posterior sd for a group whose curve does not reach its ceiling within the stimulus range (left_uni in the simulation). The HDIs still cover the truth, but the posterior means of small γ_l should not be read literally.
- **Possibly slightly undercovered:** the PSE of groups with very steep curves (slope about 6), based on a single cell (right_bi at 16,000 trials); worth checking with more replicates or other true values.
- These conclusions hold only at the one set of true values tested, and Model A assumes the curve is the same in every session. If the real sessions differ, Model A's HDIs will be too narrow, because between-session variation is treated as trial noise.

Possible follow-ups.

1. Give Model A design-based priors (in particular a lapse prior centered nearer 0.01 and a positive slope prior), with prior predictive checks, and rerun this study.
2. Draw the true parameters from the prior for each replicate (simulation-based calibration), so calibration is tested across the whole parameter range rather than at one point.
3. Check whether the right_bi PSE undercoverage at 16,000 trials holds up with more replicates.